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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC)·MG1DSCCHE100 • FUNDAMENTALS OF CHEMISTRY-I

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks:50

Part A

Multiple Choice Questions Answer All Questions

Each Question carries 1 mark

1. Which of the following is not a spectral series in the line spectrum of hydrogen? **Hund's**
2. The bond angle associated with sp2 hybridisation is **120**°
3. What type of bond is present in a water molecule? **Covalent bond**
4. What Is hyperconjugation? **Delocalization of σ electrons**
5. According to de Broglie's hypothesis, the wavelength of electron is: **inversely proportional to velocity of particle**
6. The order of acidity from strongest to weakest among these acids is: **Fluoroacetlc acid > Chloroacetic acid >Acetic acid**
7. What is the valency of sodium? **1**
8. Which of the following quantum numbers governs the spatial orientation of an atomic orbital? **Magnetic quantum number**
9. C-0,C-N, and C-H bonds are considered polar in organic molecules. Report the reason. **Wrong question give full credit**
10. Atomic radius decreases across a period due to: **increasing effective nuclear charge**

Part B

Short Answer Type Questions

Answer any 4 Questions

Each question carries 3 marks

11. **State Pauli's Exclusion Principle and provide an example**.
No two electrons in an atom can have the same four quantum number. Eg: Helium atom (Z=2), where the two electrons in 1s orbital with opposite spins (+1/2) and (-1/2), each having a unique set of quantum numbers. (1.5 + 1.5)
12. **How does electronegativity vary across a period and down a group in the periodic table? Provide reasons for these trends.**
Electronegativity increases across a period and decreases down a group. Across a period, the nuclear charge increases, pulling the electrons in the same outer

shell more strongly. Down a group, new electron shells are added, increasing the distance between the nucleus and outer electrons, which weakens the nucleus's attraction for shared electrons. (trend 1 mark + reason1+1)

13. **Discuss the significance of catenation in organic chemistry.**
    Catenation is the linkage of atoms of the same element, exemplified by carbon's ability to form long chains. This property is crucial in organic chemistry as it underpins the structure and diversity of organic compounds. (any 3 points full mark)
14. **What is effective nuclear charge (Zeff)?**
    Effective nuclear charge (Zeff) is the net positive charge that an electron experiences from the nucleus in a multi-electron atom. It is calculated as the actual nuclear charge (Z) minus the shielding constant (S), Zeff=Z-S. (1.5 + 1.5)
15. **State Hund's rule of maximum multiplicity and provide an example.**
    Hund's rule states that electrons will singly occupy each degenerate orbital within a subshell before any orbital is doubly occupied. Eg. nitrogen -$1s^2$ $2s^2$ $2p_x^1$ $2P_y^1$ $2P_z^1$. (2 + 1)
16. **Write short notes on carbanions.**
    A carbanion is a reactive intermediate with a negative charge on a carbon atom, which has three bonds and one lone pair of electrons. Formed via heterolytic bond cleavage, carbanions are powerful nucleophiles that donate electrons in reactions and can be stabilized by resonance, conjugation, and electron-withdrawing groups. (any 3 points full mark)

Part C

Short Essay Type Questions, Answer any 4 Questions
Each Question carries 7 marks

17. **Explain the limitations of Bohr theory.**
    Bohr theory's limitations include its inability to explain the spectra of atoms with more than one electron, its failure to account for the Zeeman and Stark effects (the splitting of spectral lines in magnetic and electric fields, respectively), and its contradiction of the Heisenberg Uncertainty Principle. The theory also could not explain the fine structure of spectral lines, the relative intensities of those lines, or why atoms form chemical bonds to create molecules. (any 5 points full mark)
18. **How does hyperconjugation differ from the inductive effect in stabilizing carbocations? Provide examples to explain their roles.**
    Hyperconjugation is the delocalization of sigma electrons from adjacent C-H bonds into an empty p-orbital of a carbocation, while the inductive effect is the transmission of charge through sigma bonds due to differences in electronegativity. Hyperconjugation involves the overlap of a sigma bond with a p-orbital, which distributes the positive charge, and is a more effective stabilizing mechanism for carbocations than the inductive effect, which involves a purely polar effect.
    Example: Tertiary Butyl Carbocation ($(CH_3)_3C^+$)

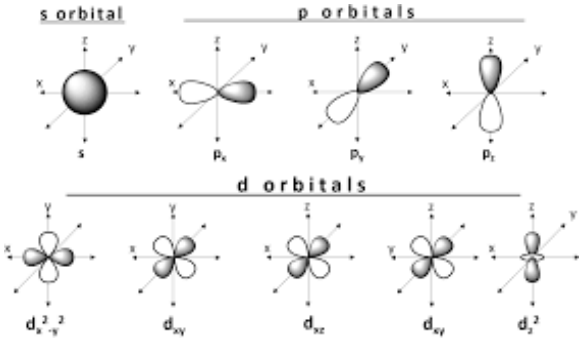
The central carbon is attached to three methyl groups. Each methyl group has three C-H bonds, adjacent to the positively charged carbon.
These σ C-H bonds overlap with the empty p-orbital of the carbocation. This allows hyperconjugation up to 9 structures can contribute. This delocalization reduces the positive charge density on the carbocation, making it much more stable. (difference 4 marks, example 3 marks)

19. Explain the differences in properties between d-block elements and those of s-block and p-block elements.

Property	s-block elements	p-block elements	d-block elements
Electronic Configuration	Outermost electrons in `ns` orbital (ns^{1-2}).	Outermost electrons in `np` orbital ($ns^2\ np^{1-6}$).	Valence electrons in both `ns` and `(n-1)d` orbitals ($(n-1)d^{1-10}\ ns^{1-2}$).
Metallicity	All are highly reactive, soft metals (except H).	Range from metals to non-metals and metalloids.	All are hard metals with high densities.
Reactivity	Very high reactivity (especially Group 1).	Diverse reactivity (contains highly reactive halogens and inert noble gases).	Less reactive and more stable than s-block metals.
Oxidation States	Exhibit only one positive oxidation state (+1 or +2).	Exhibit a wide range of oxidation states, some positive and some negative.	Exhibit multiple/variable oxidation states due to the participation of both `ns` and `(n-1)d` electrons in bonding.
Melting/Boiling Points	Low melting and boiling points compared to d-block.	Highly variable melting/boiling points depending on the element type.	High melting and boiling points due to strong metallic bonding (s and d orbital overlap).

(any 4 differences in properties full mark)

20. What is an orbital? Sketch the shapes of s, pand d orbitals.

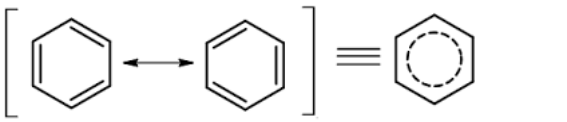


An orbital is a region around an atom's nucleus where there is a high probability of finding an electron. The shapes of s, p, and d orbitals are: s orbitals are spherical, p orbitals have a dumbbell shape, and d orbitals typically have a

cloverleaf shape. (Definition of orbital I mark, shapes s I mark, p 2 marks d 3 marks)

21. **Explain the delocalization of electrons, the concept of contributing resonance structures, and the impact of resonance on the stability of a molecule.**

Delocalization refers to the spreading of electrons rather than being confined between two atoms. This occurs in systems with conjugated double bonds or lone pairs adjacent to π bonds. Example: In benzene (C_6H_6), the six π electrons are not localized between specific carbon atoms but are delocalized across the entire ring. This leads to equal bond lengths between all carbon atoms.



Resonating Structure Resonance hybrid

Contributing resonance structures are multiple valid Lewis structures that represent different possible electron arrangements within the same molecule. Resonance stabilization lowers the overall energy of a molecule. It distributes charges and electron density more evenly. Molecules with resonance are typically less reactive and more stable than those without. (explanation 2 marks, contributing structure 3 marks, stability 2 marks)

22. **Explain the reasons behind the anomalous properties of elements like nitrogen, fluorine and oxygen within their respective groups.**

Small Atomic Size: very small atomic radii, leading to a large charge-to-radius ratio. This affects properties like ionization enthalpy and electronegativity.

High Electronegativity: highest electronegativity values, is a major factor in the formation of strong intermolecular hydrogen bonds in their hydrides, results in high melting and boiling points compared to other group member hydrides.

Absence of Vacant d-Orbitals: The valence shells of second-period elements only contain s and p orbitals (2s and 2p). The absence of vacant d-orbitals limits their maximum covalency to four (one s and three p orbitals)

Tendency to Form Multiple pπ–pπ Bonds: Due to their small size, the p-orbitals of nitrogen, oxygen, and fluorine can overlap effectively side-on to form stable pπ–pπ multiple bonds with themselves and other small elements like carbon. This results in the existence of stable diatomic molecules.

(full mark for reasonable attempt)